2025 AMC 10A Problem 24 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 24. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 24

Problem:
Call a positive integer fair if no digit is used more than once, it has no 0s, and no digit is adjacent to two greater digits. For example, 196, 23, and 12463 are fair, but 1546, 320, and 34321 are not. How many fair positive integers are there?

Answer Choices:

A. 511

B. 2584

C. 9841

D. 17711

E. 19682

Solution:

Claim. The smallest digit must appear on one of its ends.

Proof. Let S be a set of digits, and let $s = \min S$ denote the smallest element of the set.

Assume, for the sake of contradiction, that s is not positioned at either end of a fair arrangement.

Then s would have two neighbors, both of which must be greater than s, since it is the smallest digit in S.

This violates the definition of a fair arrangement, which prohibits any digit from being adjacent to two larger digits.

Therefore, the smallest digit must always occupy one of the end positions. This completes the proof of the claim.

Once the smallest digit has been placed at an end, we must verify that the remaining digits continue to form a fair arrangement.

This is achieved by repeating the same logic on the reduced set $S \setminus \{s\}$, where we again identify and place the smallest remaining digit at one of the ends.

To visualize this process, imagine constructing the arrangement step by step.

Begin with the smallest digit, then sequentially insert each next-smallest digit at either the **left end** or the **right end** of the current sequence.

If there are k digits in total, the first digit can be placed in either end position (2 options).

At each subsequent step, the next-smallest digit can again be placed on either the left or right end.

This continues until all but one digit is placed, leaving a single remaining position for the final digit.

Hence, for a set S with k digits, the total number of fair arrangements is

$$2^{k-1}.$$

Now, we will count all fair integers. Choose the k digits (no 0) in $\binom{9}{k}$ ways and arrange them fairly in 2^{k-1} ways. Summing over $k = 1, \dots, 9$,

$$T = \sum_{k=1}^{9} \binom{9}{k} 2^{k-1} = \binom{9}{1} 2^0 + \binom{9}{2} 2^1 + \cdots + \binom{9}{9} 2^8.$$

Multiply both sides by 2 and add $\binom{9}{0}$:

$$2T + \binom{9}{0} = \sum_{k=1}^{9} \binom{9}{k} 2^k + \binom{9}{0} = \sum_{k=0}^{9} \binom{9}{k} 2^k = (1+2)^9 = 3^9.$$

Thus

$$2T + 1 = 3^9 \quad \Longrightarrow \quad T = \frac{3^9 - 1}{2}.$$

Since $3^9 = 19683$, the total number of fair positive integers is

$$T = \frac{19683 - 1}{2} = \boxed{\textbf{(C) } 9841}$$